Case Study 1

Part 1

Mary Thompson is a 62-year-old retired schoolteacher who reports persistent knee pain that has gradually worsened over the past few years. The pain began about 4 years ago, initially after a day of gardening, and has progressively gotten worse. Mary experiences worse pain in the morning after getting out of bed, which somewhat improves throughout the day but intensifies again by evening. Prolonged walking, climbing stairs, and kneeling aggravate the pain, while rest and elevation provide some relief. Over-the-counter acetaminophen helps manage the discomfort. Mary's former job involved standing and walking for long periods, and although she enjoys gardening, she has limited exercise outside of this. She is slightly overweight and has no history of significant knee injuries. Currently, she only uses acetaminophen occasionally for pain relief and has never undergone any knee surgeries. Mary lives alone in a single-story house and has a supportive network of friends and family who assist with chores. Her goal is to manage her pain effectively and maintain mobility to continue gardening.

Part 2

Upon further subjective assessment, Mary has Rt knee pain and Lt knee is free, she displayed an antalgic gait and likes wearing slightly heeled shoes. Upon assessment you have noticed Mary presents with

noticeable joint swelling and warmth around both knees. There is visible muscle atrophy in the quadriceps, likely due to disuse and pain. There is tenderness over the medial and lateral aspects of the knees. No warmth or significant effusion noted in other joints. is has crepitation and limitation of flexion and extension of the knee from 10-100 on passive ROM. Pain is experienced at the end ranges of motion. Active and passive range of motion (AROM and PROM) show reduced flexibility compared to the unaffected side. Decreased strength in the quadriceps and hamstrings 2+/5 and 3-/5 respectively, with notable weakness in knee extension. Mary demonstrates an antalgic gait with a reduced stride length on the affected side. She favours the non-affected leg while walking. Difficulty performing squats and climbing stairs. Challenges with activities of daily living (ADLs) that require bending or kneeling.